

ECE 3337: Signals & Systems Analysis

Lecture 10: The Laplace Transform

**Reference: Chapter 3
(Sections 3-1 – 3-2)**

Please turn off your gadgets now

Math Moment

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$



$$\int_{-\infty}^t u(\tau) d\tau = tu(t) = r(t)$$



$$\int_{-\infty}^t r(\tau) d\tau = \frac{1}{2}t^2u(t)$$

$$\frac{du(t)}{dt} = \delta(t)$$



$$\frac{dr(t)}{dt} = u(t)$$



$$\frac{d}{dt} \left[\frac{1}{2}t^2u(t) \right] = tu(t) = r(t)$$

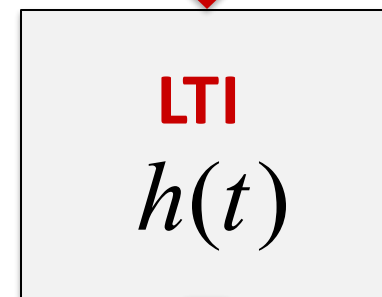


Recap: LTI Response to Sinusoids

$$\hat{\mathbf{H}}(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$

EQUIVALENT
REPRESENTATIONS

$$x(t) = A \cos(\omega t + \phi)$$



$$y(t) = A \underbrace{|\hat{\mathbf{H}}(\omega)|}_{\text{Amplitude Change}} \cos(\omega t + \phi + \underbrace{\theta}_{\text{Phase Change}})$$

Amplitude
Change

Phase
Change

$$\theta = \angle \hat{\mathbf{H}}(\omega)$$

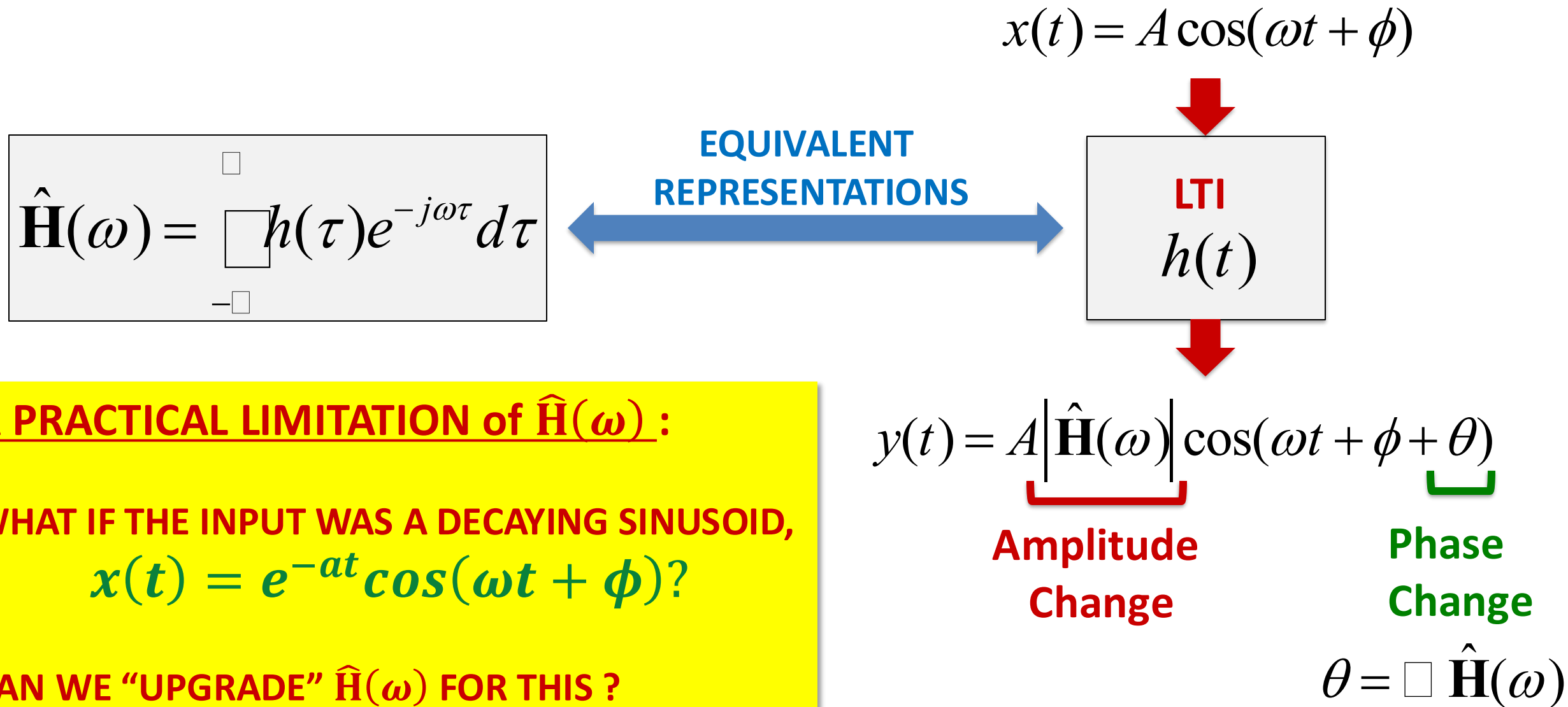
THE FREQUENCY RESPONSE FUNCTION

CONTAINS ALL OF THE INFORMATION in $h(t)$

A FULL REPRESENTATION OF THE LTI SYSTEM



Recap: LTI Response to Sinusoids



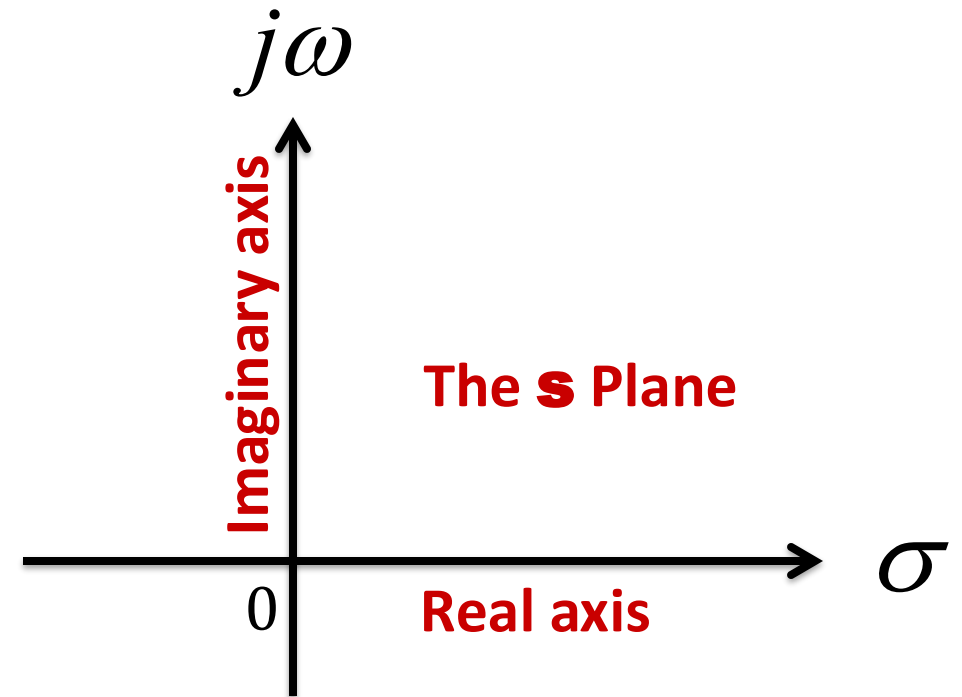


Introducing $s = \sigma + j\omega$

$$s = \sigma + j\omega$$

$\sigma = \text{Re}(s)$ describes how rapidly the signal grows or decays (attenuates)

$j\omega = \text{Im}(s)$ describes how frequently the signal oscillates



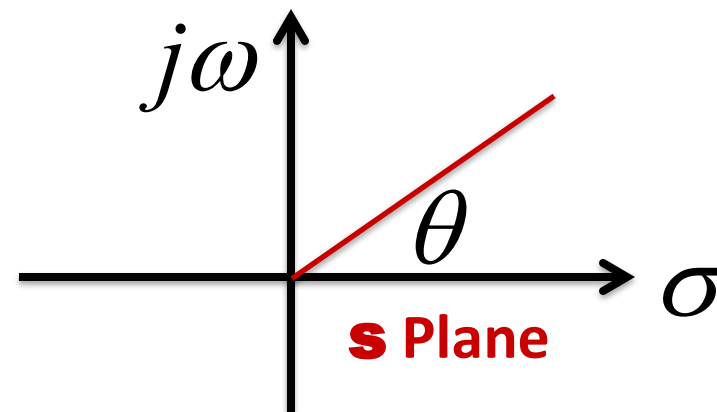
s is a two-dimensional variable



More About “**s**”: Complex Frequency

$$\mathbf{s} = \sigma + j\omega = |\mathbf{s}|e^{j\theta}$$

Complex Frequency



Physically, **s** is in the units of angular frequency, rad/sec

Both σ and ω are in frequency units

$$|\mathbf{s}| = \sqrt{\sigma^2 + \omega^2} = \text{Magnitude}$$

$$\theta = \tan^{-1}(\omega / \sigma) = \text{Phase}$$

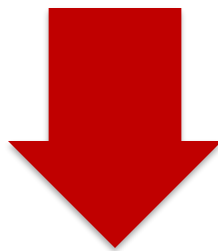
(angle calculation details not written out)



Introducing the Laplace Transform

$$\hat{\mathbf{H}}(\omega) = \int_{-\infty}^{+\infty} h(\tau) e^{-j\omega\tau} d\tau$$

Frequency Response
Function



$$\mathbf{H}(\mathbf{s}) = \int_{-\infty}^{+\infty} h(t) e^{-st} dt$$

The Laplace Transform

A general complex number that also
includes the real part σ

$$\mathbf{s} = \sigma + j\omega$$

A two-dimensional
quantity



Marquis Pierre-Simon de Laplace

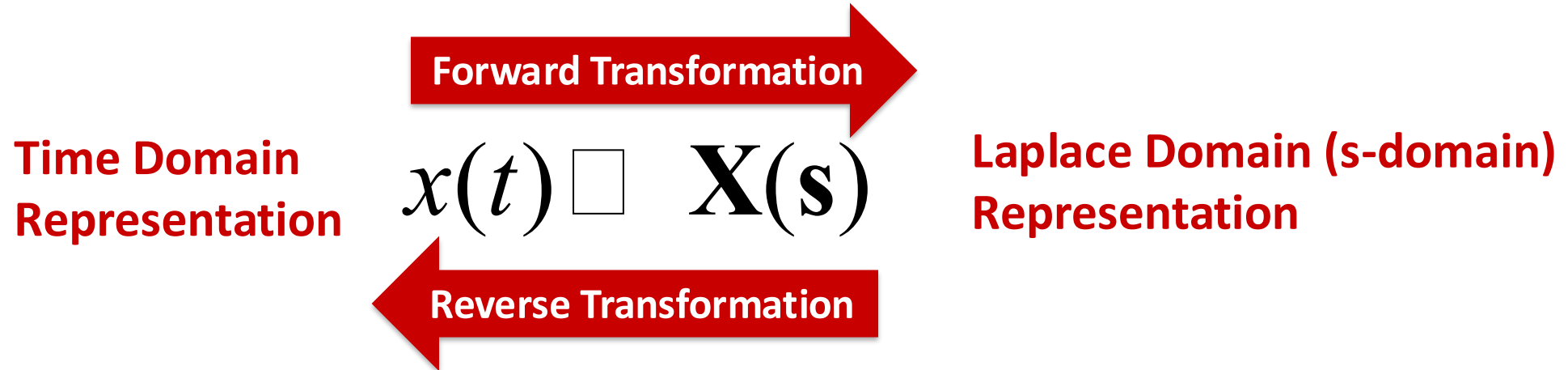
Famous French Mathematician (1749 – 1847)

- The “Isaac Newton” of France
- Published the now famous Laplace Transform in 1779 for solving differential equations
- His ideas were developed into their modern form after World War II
- **Major contribution that makes our work much easier!**
 - **Converts linear differential equations into polynomial problems, convolutions into multiplications, and initial conditions a breeze**





The Complex Frequency Domain



Signals and systems can be expressed in the two-dimensional s domain just as easily and completely as in the time domain

- We can go back and forth as needed
- The two-dimensional nature of the s -domain allows more comprehensive analysis
- Once we get used to it, we will find this domain to be more convenient for day-to-day engineering work compared to the time domain



The Frequency Domain Offers Benefits

Interesting Property: convolutions in the time domain become multiplications in the frequency domain

$$x(t) \square \mathbf{X(s)}$$

$$x(t) * h(t) \square \mathbf{X(s)H(s)}$$

It can be easier to convert signals to the s-domain, multiply them, and transform the product back to the time domain!



Other Benefits of the Laplace Domain

We will learn that the Laplace method will allow us to avoid solving differential equations by integration

- They will become polynomials in the s -domain, so easier to work with!**
- The Laplace transform was originally invented to help solve differential equations**

We will also learn that the Laplace method will allow us to handle complicated initial conditions easily

Bottom Line: There is a lot of gain from what we are about to learn!

- It will feel abstract at first, but hang in there...***

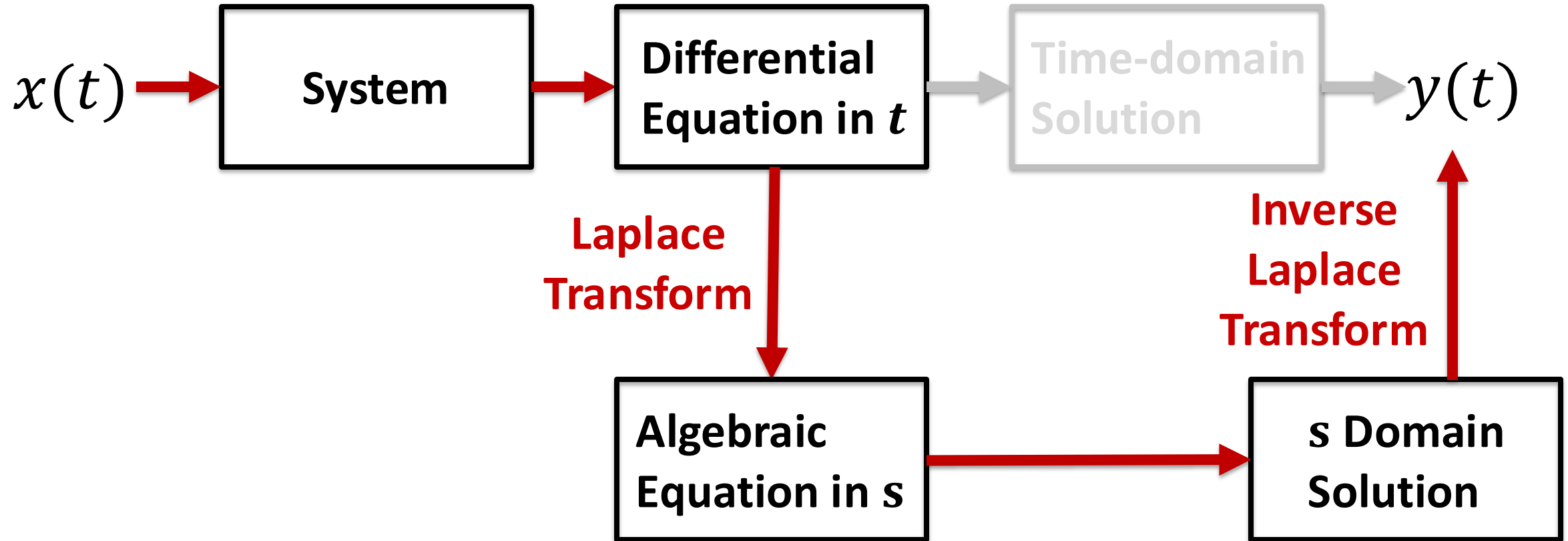


Our New Parallel Universe!



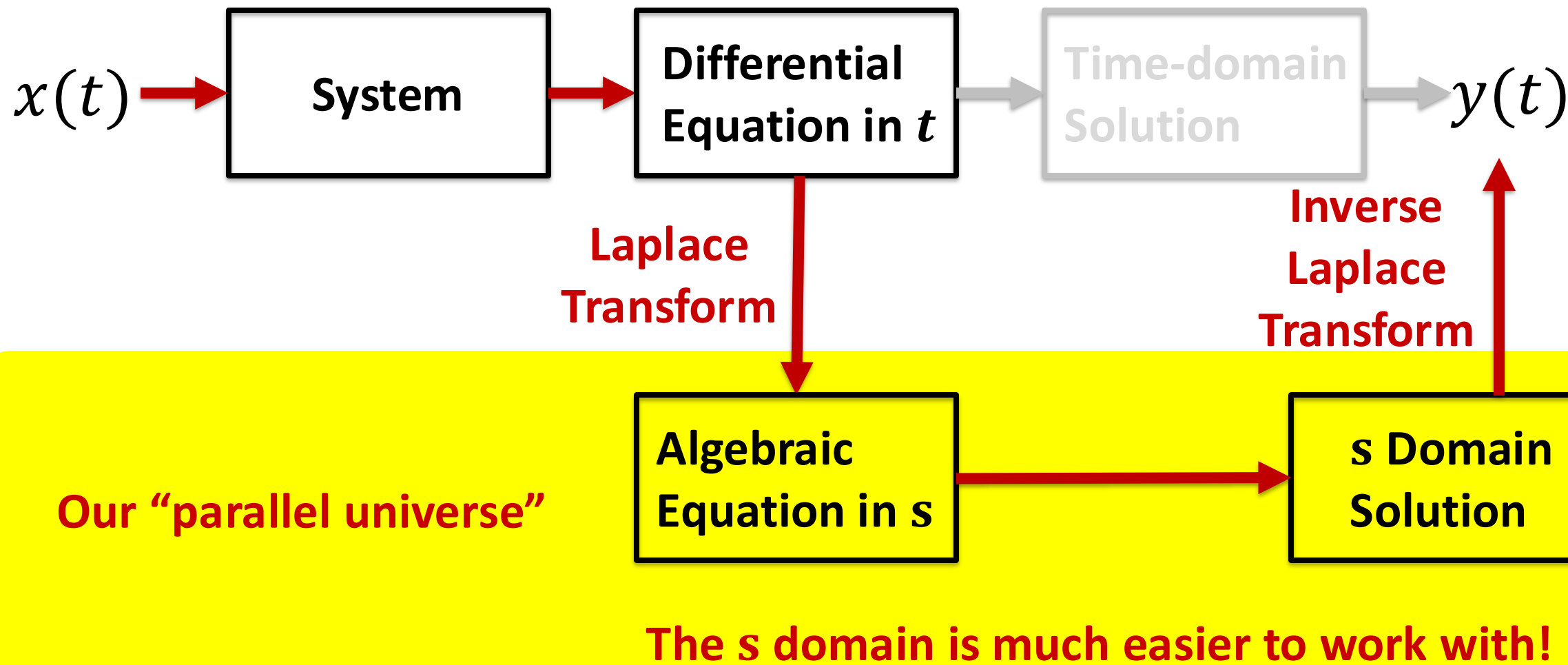


An Alternative Way





Our “Parallel Universe”





The One-sided Laplace Transform

$$\mathbf{H(s)} = \int_{-\infty}^{+\infty} h(t)e^{-st} dt$$

Full (bilateral)
Laplace Transform
(general)

$$\mathbf{H(s)} = \int_{0-}^{+\infty} h(t)e^{-st} dt$$


One-sided (unilateral)
Laplace Transform
(for causal signals/systems)

Needed to include the effect of an impulse input at the origin, and any initial conditions at $t = 0 -$

The unilateral version of the Laplace transform is suited for handling causal signals and systems – what we are most interested in



The Inverse Laplace Transform

The inverse Laplace transform is required to translate a signal/system from the s-domain back to the time domain

$$\mathcal{L}^{-1}[\mathbf{X}(s)] = x(t)$$

Notation  $= \frac{1}{j2\pi} \int_{\sigma-j\infty}^{\sigma+j\infty} \mathbf{X}(s)e^{st} ds$ **Integration in the complex plane**

Intuitively, this formula is saying that a given signal $x(t)$ can be written as a weighted summation of complex exponentials e^{st}

Practical challenges:

1. Calculating the inverse Laplace transform requires knowledge of complex-domain calculus (outside the scope of ECE 3337)
2. The inverse transform may not be unique



Fortunately ...

In ECE 3337, we will restrict ourselves to the unilateral Laplace transform (limited to causal signals and systems)

$$\mathbf{X(s)} = \int\limits_{0-}^{+\infty} x(t)e^{-st} dt = \mathcal{L}[x(t)]$$

For this special case,

- The inverse transform is unique, so can be obtained from a table
- We can get by without heavy-duty complex calculus (integration in the complex plane, regions of convergence, paths of convergence, etc.)
- Limitation: we can only analyze causal, physically realizable systems



Laplace Transform Tables

We will rely on tables of Laplace transforms for the most used signals

- This method works because of the uniqueness of the inverse for unilateral transforms
- Most signals of interest can be expressed in terms of these table entries
- Some of these table entries will become memorable
- We will learn to recognize useful patterns among these entries
- Can you identify some?

Table 3-2: Examples of Laplace transform pairs. Note that $x(t) = 0$ for $t < 0^-$ and $T \geq 0$.

Laplace Transform Pairs		
	$x(t)$	$X(s) = \mathcal{L}[x(t)]$
1	$\delta(t)$	1
1a	$\delta(t - T)$	e^{-Ts}
2	$u(t)$	$\frac{1}{s}$ 
2a	$u(t - T)$	$\frac{e^{-Ts}}{s}$
3	$e^{-at} u(t)$	$\frac{1}{s + a}$
3a	$e^{-a(t-T)} u(t - T)$	$\frac{e^{-Ts}}{s + a}$
4	$t u(t)$	$\frac{1}{s^2}$ 
4a	$(t - T) u(t - T)$	$\frac{e^{-Ts}}{s^2}$
5	$t^2 u(t)$	$\frac{2}{s^3}$ 
6	$t e^{-at} u(t)$	$\frac{1}{(s + a)^2}$
7	$t^2 e^{-at} u(t)$	$\frac{2}{(s + a)^3}$
8	$t^{n-1} e^{-at} u(t)$	$\frac{(n-1)!}{(s + a)^n}$
9	$\sin(\omega_0 t) u(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$
10	$\sin(\omega_0 t + \theta) u(t)$	$\frac{s \sin \theta + \omega_0 \cos \theta}{s^2 + \omega_0^2}$
11	$\cos(\omega_0 t) u(t)$	$\frac{s}{s^2 + \omega_0^2}$



Laplace Table of Properties

A similar thing will happen with properties

- Many complicated operations in the time domain will have simpler **S**-domain versions (or *vice versa*)
- With our handy tables, we will be able to handle most situations
- Many of these properties will also become memorable

They better!

Table 3-1: Properties of the Laplace transform for causal functions; i.e., $x(t) = 0$ for $t < 0$.

Property	$x(t)$	$\mathbf{X(s)} = \mathcal{L}[x(t)]$
1. Multiplication by constant	$K x(t)$	$\longleftrightarrow K \mathbf{X(s)}$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t)$	$\longleftrightarrow K_1 \mathbf{X_1(s)} + K_2 \mathbf{X_2(s)}$
3. Time scaling	$x(at), \quad a > 0$	$\longleftrightarrow \frac{1}{a} \mathbf{X}\left(\frac{s}{a}\right)$
4. Time shift	$x(t - T) u(t - T)$	$\longleftrightarrow e^{-Ts} \mathbf{X(s)}, \quad T \geq 0$
5. Frequency shift	$e^{-at} x(t)$	$\longleftrightarrow \mathbf{X(s + a)}$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$\longleftrightarrow s \mathbf{X(s)} - x(0^-)$
7. Time 2nd derivative	$x'' = \frac{d^2x}{dt^2}$	$\longleftrightarrow s^2 \mathbf{X(s)} - sx(0^-) - x'(0^-)$
8. Time integral	$\int_{0^-}^t x(t') dt'$	$\longleftrightarrow \frac{1}{s} \mathbf{X(s)}$
9. Frequency derivative	$t x(t)$	$\longleftrightarrow -\frac{d}{ds} \mathbf{X(s)} = -\mathbf{X'(s)}$
10. Frequency integral	$\frac{x(t)}{t}$	$\longleftrightarrow \int_s^\infty \mathbf{X(s')} ds'$
11. Initial value	$x(0^+)$	$= \lim_{s \rightarrow \infty} s \mathbf{X(s)}$
12. Final value	$\lim_{t \rightarrow \infty} x(t) = x(\infty)$	$= \lim_{s \rightarrow 0} s \mathbf{X(s)}$
13. Convolution	$x_1(t) * x_2(t)$	$\longleftrightarrow \mathbf{X_1(s)} \mathbf{X_2(s)}$



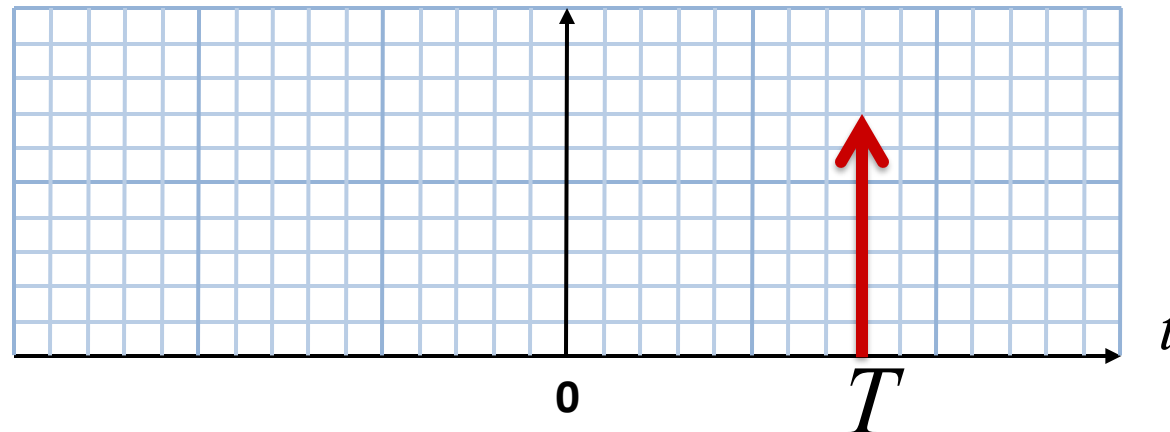
Laplace Transform of the Impulse

Impulse

$$\mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t) e^{-st} dt = 1$$

Delayed Impulse

$$\mathcal{L}[\delta(t - T)] = \int_{0^-}^{+\infty} \delta(t - T) e^{-st} dt = e^{-sT}$$





The Unit Step

Unit Step

$$\mathcal{L}[u(t)] = \int_{0-}^{+\infty} u(t)e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_{0-}^{\infty} = -\frac{1}{s} (0 - 1) = \frac{1}{s}$$

Delayed Unit Step

$$\begin{aligned} \mathcal{L}[u(t - T)] &= \int_{0-}^{+\infty} u(t - T)e^{-st} dt \\ &= \int_T^{+\infty} e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_T^{\infty} = \frac{1}{s} e^{-Ts} \end{aligned}$$

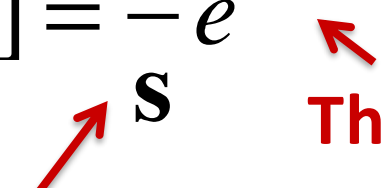


Do You See the Pattern?

The delayed unit step can be thought of as a unit step convolved with the delayed impulse:

$$u(t - T) = u(t) * \delta(t - T)$$

Look at this in the s-domain:

$$\mathcal{L}[u(t - T)] = \frac{1}{s} e^{-Ts}$$


The “delayer” term

Laplace Transform of the unit step

4. Time shift

$$x(t - T) u(t - T), T \geq 0 \quad \longleftrightarrow \quad e^{-Ts} X(s)$$



Similarly, ...

The delayed impulse can be revisited similarly:

$$\mathcal{L}[\delta(t - T)] = 1 \cdot e^{-sT}$$

Laplace
Transform of
unit impulse



The “delayer”





Shift Property

It is easy to see that if: $x(t)u(t) \square \mathbf{X}(s)$

Then, $\mathcal{L}[x(t - T)u(t - T)] = \int_{0-}^{\infty} x(t - T)u(t - T)e^{-st} dt$

Let $\mu = t - T$

$$= \int_{0-}^{\infty} x(\mu)u(\mu)e^{-s(\mu+T)} d\mu$$
$$= e^{-sT} \int_0^{\infty} x(\mu)e^{-s\mu} d\mu = e^{-sT} \mathbf{X}(s)$$

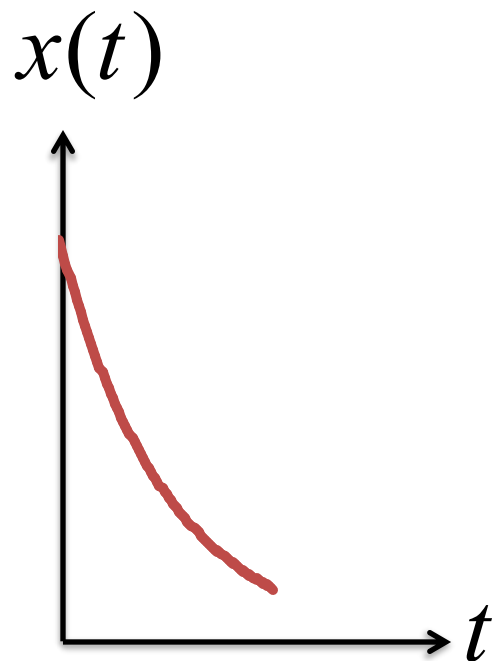


This only works for $T > 0$ since the signal may no longer be causal for a negative T

The “delayer”



The Decaying Exponential



$$x(t) = e^{-at} u(t)$$

**Multiplying by the unit step
forces it to be causal**

$$\mathbf{X(s)} = \int_{0-}^{\infty} e^{-at} u(t) e^{-st} dt = \int_{0-}^{\infty} e^{-(s+a)t} dt$$

$$= \left. \frac{e^{-(s+a)t}}{-(s+a)} \right|_0^{\infty} = 0 - \frac{1}{-(s+a)} = \frac{1}{(s+a)}$$

How does this relate to the RC circuit?

**Memorable Transform
(see the table)**



The Sinusoid

$$\begin{aligned}x(t) &= \cos(\omega_0 t)u(t) = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})u(t) \\ \Rightarrow \mathbf{X}(\mathbf{s}) &= \frac{1}{2}\{\mathcal{L}[e^{j\omega_0 t}u(t)] + \mathcal{L}[e^{-j\omega_0 t}u(t)]\} \\ &= \frac{1}{2}\left\{\frac{1}{\mathbf{s} - j\omega_0} + \frac{1}{\mathbf{s} + j\omega_0}\right\} \\ &= \frac{1}{2}\left\{\frac{(\mathbf{s} + \cancel{j\omega_0}) + (\mathbf{s} - \cancel{j\omega_0})}{\mathbf{s}^2 + \omega_0^2}\right\} = \frac{\mathbf{s}}{\mathbf{s}^2 + \omega_0^2}\end{aligned}$$

Yet another classic worth memorizing



Watch the Denominators!

We should always take a close look at denominators, *every time we encounter them*

- **Bad things happen when a denominator goes to zero**

$$\mathcal{L}[\cos(\omega_0 t)u(t)] = \frac{s}{s^2 + \omega_0^2} = \frac{s}{(s - j\omega_0)(s + j\omega_0)}$$

This denominator goes to zero under either of the following two conditions:

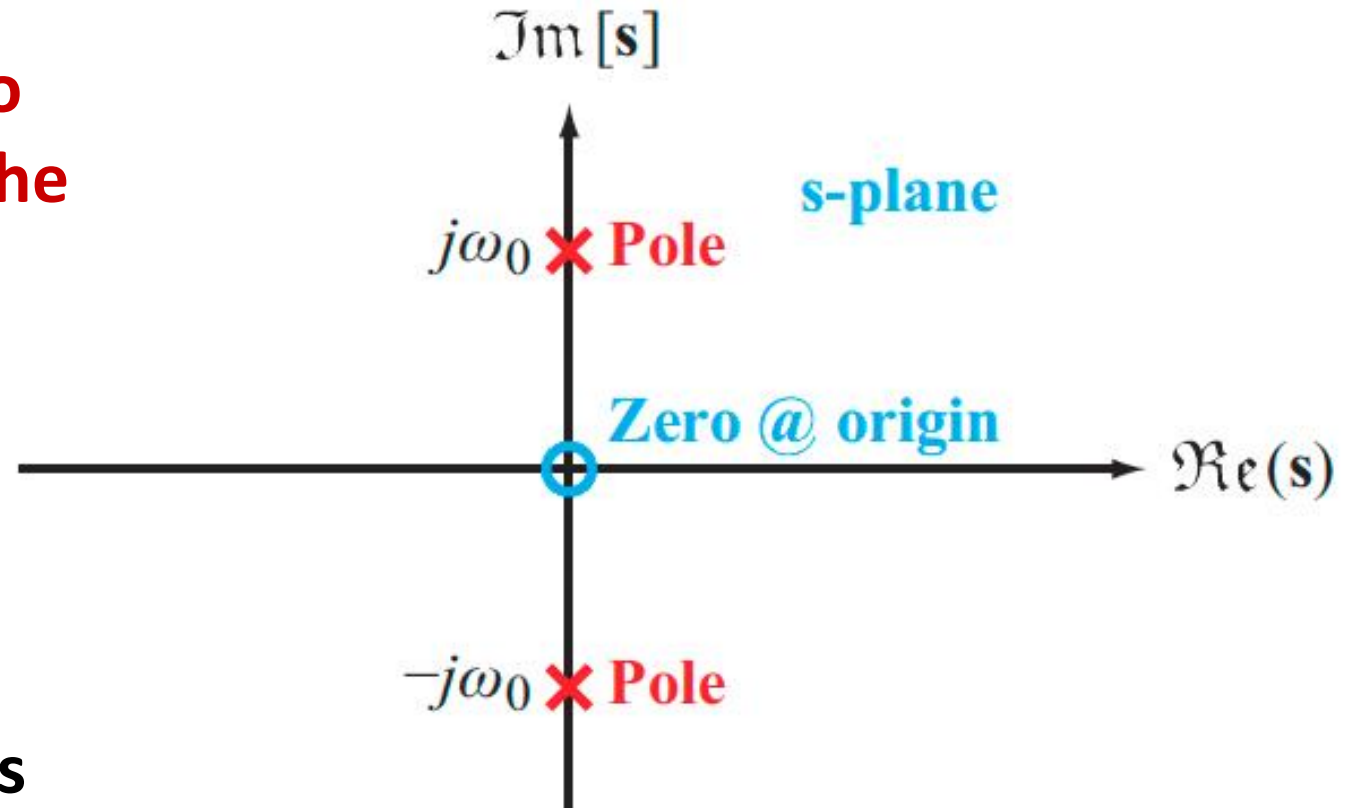
$$s = j\omega_0$$

$$s = -j\omega_0$$



Poles & Zeros

- Points in the complex **s**-plane where the denominator can go to zero are called “poles” because the response is infinite
- Zeros are points where the numerator can go to zero
- For this example, we have two poles and one zero
- The poles are complex conjugates of each other for our example



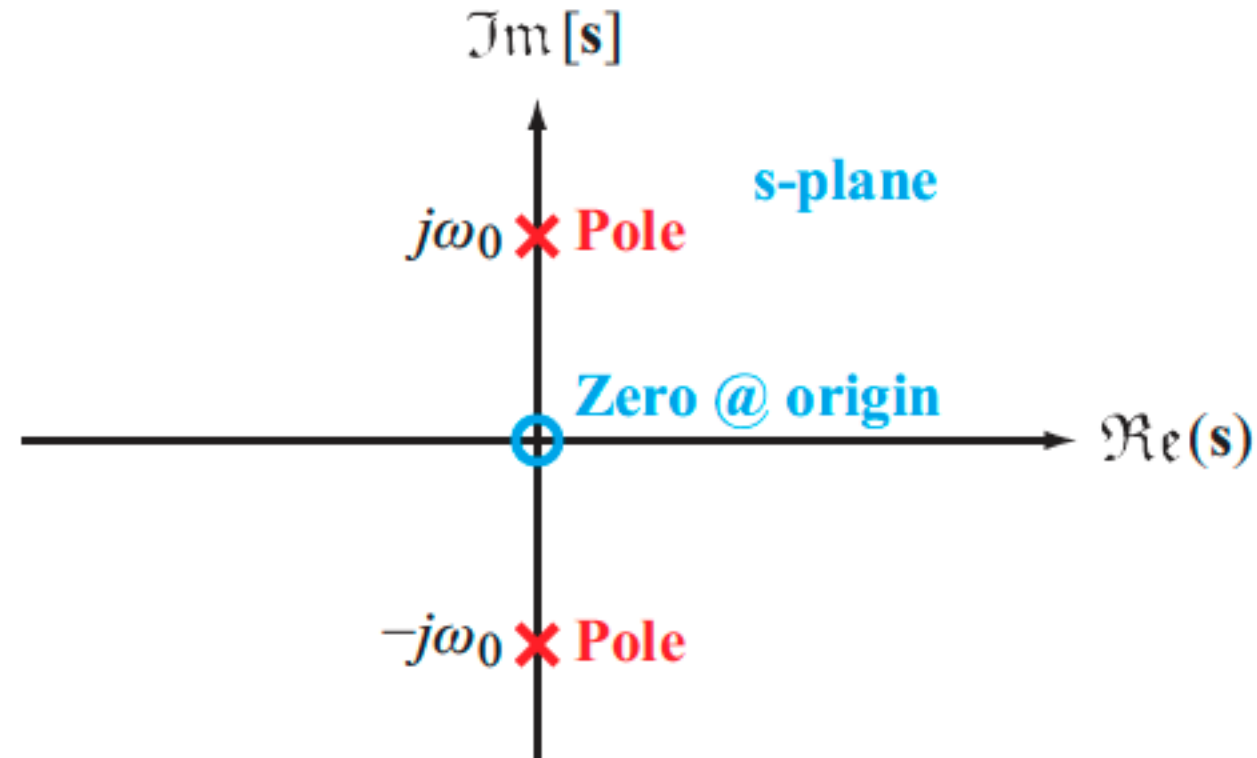
$$\mathcal{L}[\cos(\omega_0 t)u(t)] = \frac{s}{(s - j\omega_0)(s + j\omega_0)}$$



Pole – Zero Plots

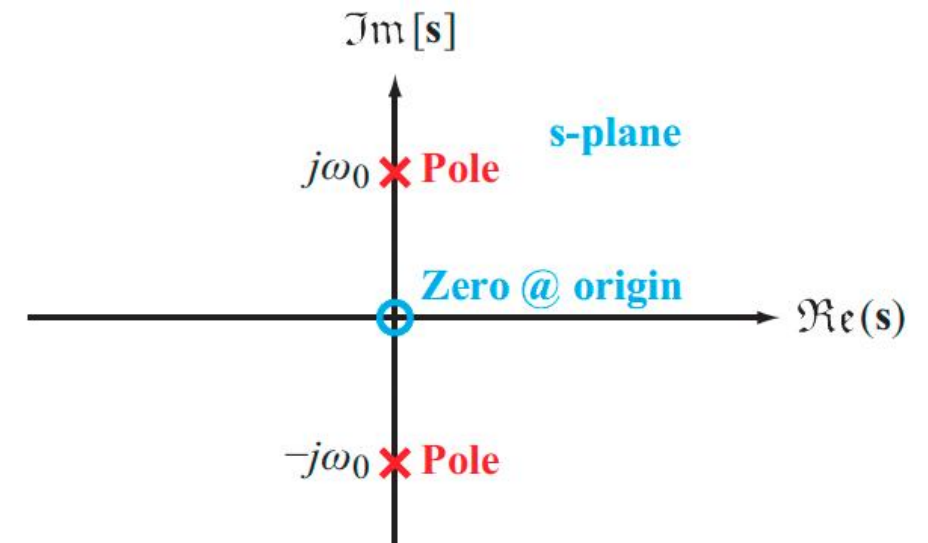
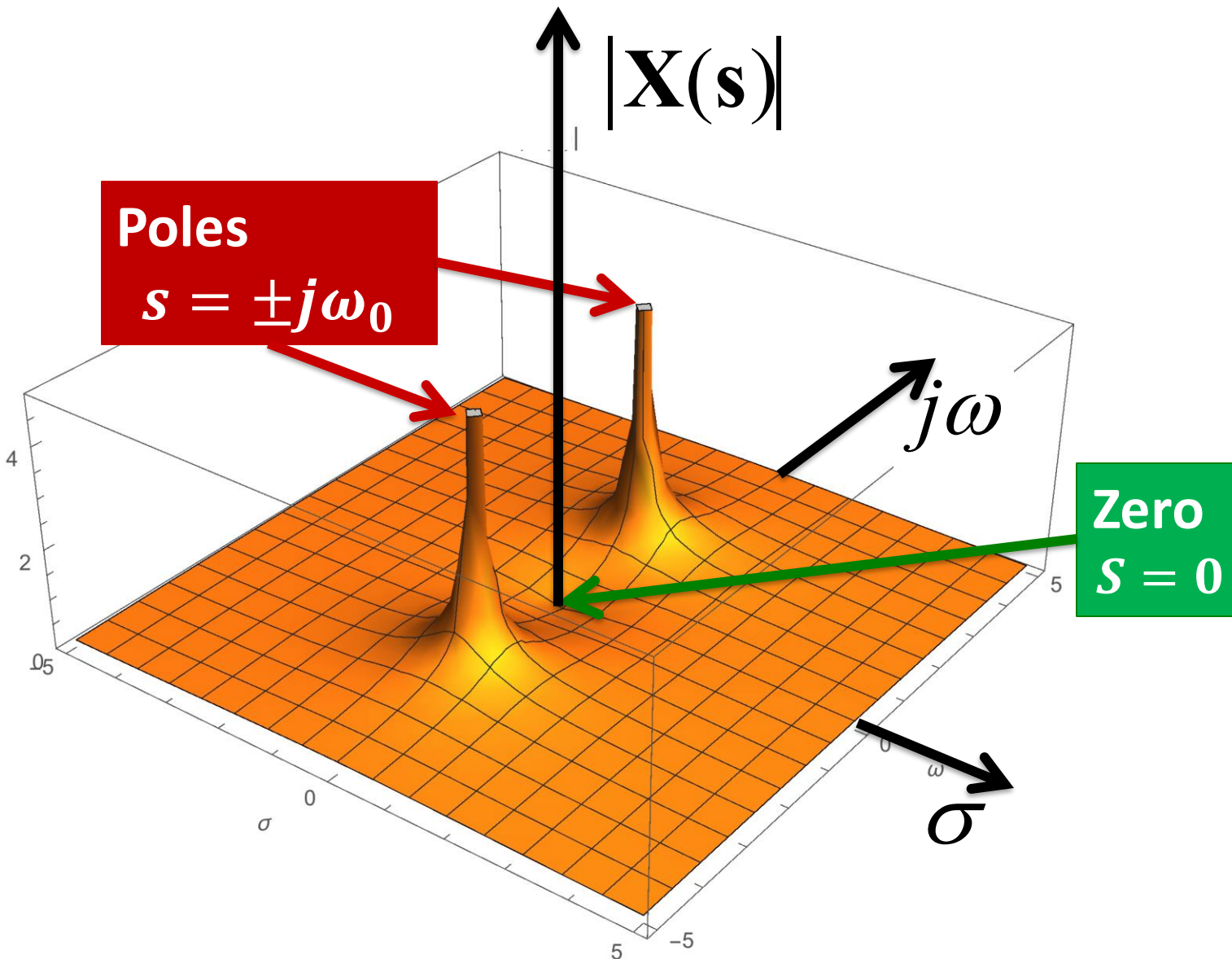
- Plots like this are called “pole-zero plots”
- The poles and zeros together represent a surprisingly complete description of a system/signal
- They can often be written as a ratio of polynomials:

$$\mathbf{X}(s) = \frac{\mathbf{N}(s)}{\mathbf{D}(s)}$$



Once we become familiar with these plots, we will be able to visually interpret the system's behavior!

3-D Pole – Zero Plot

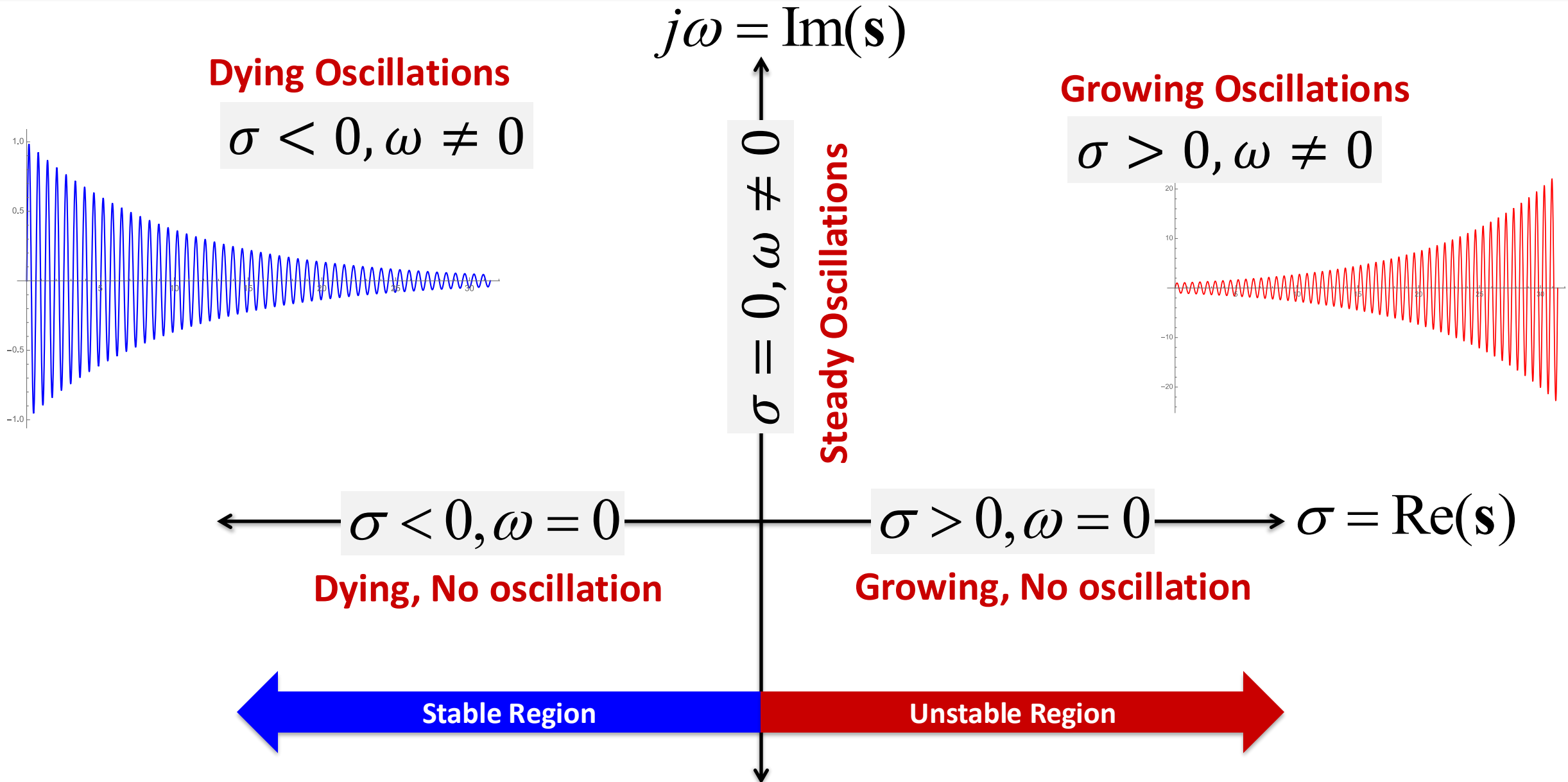


$$X(s) = \frac{s}{s^2 + \omega_0^2}$$

$$= \frac{s}{(s + j\omega_0)(s - j\omega_0)}$$



How Locations of Poles Define Behaviors





Classroom Activity

Following the same steps, compute the Laplace Transform for the following signal:

$$x(t) = \sin(\omega_0 t)u(t)$$

$$\mathbf{X(s)} = \frac{\omega_0}{s^2 + \omega_0^2}$$

**Make sure to arrive
at this Answer**

How about the signal $x(t) = e^{-at} \sin(\omega_0 t)u(t)$?

Do you notice a pattern here?



Summary

The Laplace transform is our window into a “parallel universe” that provides important advantages

$$\mathbf{X(s)} = \int\limits_{0-}^{+\infty} x(t)e^{-st} dt = \mathcal{L}[x(t)]$$

We will restrict attention to causal signals and systems in this course, and this makes the use of this transform easier by avoiding complex calculus

– In other words, the unilateral transform only in ECE 3337



Summary: Laplace Tables are our new BFF

Table 3-2: Examples of Laplace transform pairs. Note that $x(t) = 0$ for $t < 0^-$ and $T \geq 0$.

Laplace Transform Pairs			
	$x(t)$		$\mathbf{X}(s) = \mathcal{L}[x(t)]$
1	$\delta(t)$	$\longleftrightarrow$	1
1a	$\delta(t - T)$	$\longleftrightarrow$	e^{-Ts}
2	$u(t)$	$\longleftrightarrow$	$\frac{1}{s}$
2a	$u(t - T)$	$\longleftrightarrow$	$\frac{e^{-Ts}}{s}$
3	$e^{-at} u(t)$	$\longleftrightarrow$	$\frac{1}{s + a}$
3a	$e^{-a(t-T)} u(t - T)$	$\longleftrightarrow$	$\frac{e^{-Ts}}{s + a}$
4	$t u(t)$	$\longleftrightarrow$	$\frac{1}{s^2}$
4a	$(t - T) u(t - T)$	$\longleftrightarrow$	$\frac{e^{-Ts}}{s^2}$
5	$t^2 u(t)$	$\longleftrightarrow$	$\frac{2}{s^3}$
6	$te^{-at} u(t)$	$\longleftrightarrow$	$\frac{1}{(s + a)^2}$

**Keep 'em Handy
always!
Memorize key entries
Remember the patterns**



For the Next Class

Work through the textbook examples in Sections 3.1 – 3.2

- Try them repeatedly until you can do them correctly and quickly without help

Study Section 3.3 & 3.4

Bring the Laplace Tables to each class from now onwards

How to Study:

1. Read the assigned sections to understand the basic concepts, and memorize them using flashcards, crib sheets, ... whatever
2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test
3. Write down your questions and bring them to class